Math 3IA3 - Assignment 1

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Contents

Exercise 1.1.15

Show that if $f : A \to B$ and G and H are both subsets of B, then:

(a) $f^{-1}(G \cup H) = f^{-1}(G) \cup f{-}1(H)$.

Solution:

We will prove that $f^{-1}(G \cup H) = f^{-1}(G) \cup f^{-1}(H)$ by first showing that $f^{-1}(G) \cup f^{-1}(H) \subseteq f^{-1}(G \cup H)$ and that $f^{-1}(G \cup H) \subseteq f^{-1}(G) \cup f^{-1}(H)$.

(I) Show that $f^{-1}(G) \cup f^{-1}(H) \subseteq f^{-1}(G \cup H)$.

Let $b \in (f^{-1}(G) \cup f^{-1}(H))$. Then, by definition of inverse, we know $f(b) \in G \vee f(b) \in H$. Thus, $f(b) \in G \cup H$. Taking the inverse of both sides, yields $b \in f^{-1}(G \cup H)$. Thus, since $b \in (f^{-1}(G) \cup f^{-1}(H)) \implies b \in f^{-1}(G \cup H)$, $(f^{-1}(G) \cup f^{-1}(H)) \subseteq f^{-1}(G \cup H)$.

(II) Show that $f^{-1}(G \cup H) \subseteq f^{-1}(G) \cup f^{-1}(H)$.

Let $b \in f^{-1}(G \cup H)$. Applying f to both sides yields $f(b) \in (G \cup H) \implies f(b) \in G \vee f(b) \in H$. Thus, reapplying the inverse we have that $b \in f^{-1}(G) \vee \cup f^{-1}(H)$, which, by definition of inverse implies that $b \in f^{-1}(G) \cup f^{-1}(H)$. Thus, since $b \in f^{-1}(G \cup H) \implies b \in (f^{-1}(G) \cup f^{-1}(H))$, $f^{-1}(G \cup H) \subseteq (f^{-1}(G) \cup f^{-1}(H))$.

Therefore, since $f^{-1}(G) \cup f^{-1}(H) \subseteq f^{-1}(G \cup H)$ and $f^{-1}(G \cup H) \subseteq f^{-1}(G) \cup f^{-1}(H)$, we necessarily have that $f^{-1}(G \cup H) = f^{-1}(G) \cup f^{-1}(H)$.

(b) $f^{-1}(G \cap H) = f^{-1}(G) \cap f^{-1}(H)$.

Solution:

We will likewise prove that $f^{-1}(G \cap H) = f^{-1}(G) \cap f^{-1}(H)$ by first showing that $f^{-1}(G \cap H) \subseteq (f^{-1}(G) \cap f^{-1}(H))$ and that $(f^{-1}(G) \cap f^{-1}(H)) \subseteq f^{-1}(G \cap H)$.

(I) Show that $f^{-1}(G \cap H) \subseteq (f^{-1}(G) \cap f^{-1}(H))$.

Let $b \in f^{-1}(G \cap H)$. Therefore, applying the function to both sides, we know that $f(b) \in (G \cap H) \implies f(b) \in G \wedge f(b) \in H$. Thus, applying the inverse, we have $b \in f^{-1}(G) \wedge b \in f^{-1}(H)$ and thus, by definition of intersection, we have $b \in f^{-1}(G) \cap f^{-1}(H)$. Thus, since $b \in f^{-1}(G \cap H) \implies b \in (f^{-1}(G) \cap f^{-1}(H))$, $f^{-1}(G \cap H) \subseteq (f^{-1}(G) \cap f^{-1}(H))$.

(II) $(f^{-1}(G) \cap f^{-1}(H)) \subseteq f^{-1}(G \cap H)$.

Let $b \in f^{-1}(G) \cap f^{-1}(H)$. Therefore, applying the function to both sides, we know that

$f(b) \in G \wedge f(b) \in H$. Thus, by definition of inverse, we know that $f(b) \in (G \cap H)$, and by reapplying the inverse to both sides, we yield $b \in f^{-1}(G \cap H)$. Thus, since $b \in (f^{-1}(G) \cap f^{-1}(H)) \implies b \in f^{-1}(G \cap H)$, $(f^{-1}(G) \cap f^{-1}(H)) \subseteq f^{-1}(G \cap H)$.

Therefore, since $f^{-1}(G \cap H) \subseteq (f^{-1}(G) \cap f^{-1}(H))$ and $(f^{-1}(G) \cap f^{-1}(H)) \subseteq f^{-1}(G \cap H)$ we necessarily have that $(f^{-1}(G) \cap f^{-1}(H)) = f^{-1}(G \cap H)$